"CYBER SECURITY ISSUES IN INDIA" -- INDIA'S APPROACH TO CYBER SECURITY AND NATIONAL CYBER SECURITY

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Abstract—
The rapid growth of digital technologies, internet usage, and e-governance in India has significantly increased the importance of cyber security at both national and organizational levels. With the expansion of digital infrastructure, India faces a wide range of cyber security challenges including cybercrime, cyber terrorism, cyber espionage, and cyber warfare. This paper examines major cyber security issues in India and evaluates the country's approach towards strengthening national cyber security frameworks. The study is based on both primary and secondary data, including survey analysis and insights from existing research on global cyber threats, risk management, and emerging technological vulnerabilities. Findings indicate that the rise in digital transactions, cloud computing, and mobile networks has increased exposure to cyber risks, leading to financial losses, data breaches, and threats to critical infrastructure. Additionally, lack of standardized cyber risk data and limited awareness among users remain significant challenges. The study highlights that while India has implemented initiatives such as the Information Technology Act and national cyber security policies, gaps still exist in enforcement, awareness, and technological preparedness. This research concludes that a comprehensive approach involving stronger legal frameworks, improved public awareness, advanced security technologies, and collaboration between government and private sectors is essential to enhance India's national cyber security ecosystem. [1], [2]

Keywords— Cyber Security, Cyber Crime, National Cyber Security, India, Cyber Laws, Data Protection

I. INTRODUCTION

Over the years, Information Technology has transformed the global economy and connected people and markets in ways beyond imagination. With Information Technology gaining the center stage, nations across the world are experimenting with innovative ideas for economic development and inclusive growth. An increasing proportion of the world's population is migrating to cyberspace to communicate, enjoy, learn, and conduct commerce. While this transformation has improved efficiency, connectivity, and accessibility, it has also increased vulnerabilities to cyber threats. [3] The dependence on digital infrastructure has made sensitive data, financial systems, and critical information infrastructures more vulnerable to cyber attacks.

Cyber security means the protection of computer systems, networks, and data from unauthorized access, attacks, and damage. Cyber threats such as cybercrime, cyber terrorism, cyber espionage, and cyber warfare are growing rapidly. These threats not only result in financial losses but also cause serious risks to the security of the nation and individual privacy. [4] The increasing use of mobile devices, social media platforms, and online services has further expanded the attack surface for cyber criminals.

The origin of a disruption, the identity of the perpetrator or the motivation for it can be difficult to ascertain and the act can take place from virtually anywhere. These attributes facilitate the use of Information Technology for disruptive activities. As such, cyber security threats pose one of the most serious economic and national security challenges Government initiatives such as Digital India and the expansion of online services have significantly boosted digital inclusion, but they have also exposed gaps in cyber security awareness and infrastructure. [5] Cyber attacks targeting banking systems, government databases, and critical infrastructure highlight the urgent need for a robust cyber security framework.

The Government of India has taken several steps to address cyber security issues, including the implementation of the Information Technology Act, 2000, and the formulation of the National Cyber Security Policy. [6] However, challenges such as lack of awareness, shortage of skilled professionals, weak enforcement of laws, and evolving nature of cyber threats continue to create challenges for cyber security management.

This paper explores cyber security issues in India and reviews how the country is working to strengthen its national cyber security. It also focuses on understanding different types of cyber threats, existing legal frameworks, and the need for stronger strategies to ensure a secure digital environment. [7]

II. LITERATURE REVIEW

As digital technologies grow and people depend more on cyberspace, cyber security has become a key research area. Researchers have studied cybercrime trends, risk management, and how effective security measures are across different countries.

"Kuzior et al. (2024) researched global cybercrime trends and found that cyber threats have been increasing quickly due to digital transformation, higher internet usage, and the growth of e-commerce and digital payment systems [8]. The study also showed that cybercrime has increased in recent years, especially after the COVID-19 pandemic, with ransomware, phishing, and data breaches being among the most common threats. It further noted that such attacks can lead to serious financial losses and can also disrupt critical infrastructure, making cyber security an important concern for governments around the world."

Cremer et al. (2022) carried out a systematic review on cyber risk and the availability of cybersecurity data [9]. According to the study, cyber security research and risk management are often limited by the unavailability of standardized and publicly accessible data . It explained that limited data makes it harder for organizations and policymakers to properly assess risks and apply effective security measures. The authors also noted that cyber insurance and data-driven approaches play an important role in dealing with cyber threats.

Many studies have highlighted the importance of human factors in cyber security. Research shows that low awareness, weak security practices, and simple human mistakes often lead to cyber incidents. [11] Social engineering attacks, phishing, and identity theft often exploit human vulnerabilities rather than technical weaknesses, making user education an essential component of cyber security.

In India, studies show that despite significant progress in digitalization, the country still struggles with issues like inadequate infrastructure, a lack of skilled professionals, and weak enforcement of cyber laws [12]. Although acts such as the Information Technology Act, 2000, and the National Cyber Security Policy have laid the groundwork for cyber governance, continuous improvements are needed to deal with evolving threats

Overall, the literature shows that cyber security keeps evolving and needs a balanced approach involving technology, law, and organizational efforts . As cyber threats become more complex, along with issues like low awareness and limited data, there is a clear need for a more comprehensive and collaborative approach to improve cyber security at both national and global levels [8]–[10].

III. METHODOLOGY

This research paper adopts a combination of both primary and secondary research methods to analyze cyber security issues in India and evaluate the country's approach toward national cyber security.

A. Research Design

This research takes a descriptive and analytical approach, focusing on key cyber security challenges, user awareness, and existing policies and frameworks in India. It aims to give an overall understanding of cyber threats and how they affect individuals and organizations

B. Data Collection

Primary data were gathered through a structured questionnaire to understand cyber security awareness, common threats faced by users, and the safety measures they follow. The survey covered 50 respondents from Hyderabad and Secunderabad, including students, working professionals, and general internet users to ensure a mix of responses.

Secondary data were collected from various reliable sources such as books , academic journals, research papers , government reports, newspapers, and online resources. [13] These sources provided insights into global cyber security trends, cyber risk management, and India's cyber laws and policies.

C. Data Analysis Techniques

The collected data were examined using simple statistical methods like tables, percentages, and graphs. The questionnaire responses were organized and interpreted to find patterns in users' awareness of cyber security, the types of threats they face, and the preventive measures they use.

D. Limitations of the Study

With only 50 respondents, the study may not completely represent the overall population of India. Part of the data is self-reported, which means it may contain some bias or inaccuracies. Since cyber threats keep evolving, it becomes difficult for one study to fully capture all the latest trends

IV. TYPES OF CYBER SECURITY THREATS

As we grow more dependent on the Internet for our daily activities, we also become more vulnerable to any disruptions caused in and through cyberspace. The rapidity with which this sector has grown has meant that governments and private companies are still trying to figure out both the scope and meaning of security in cyberspace and apportioning responsibility. In India, individuals, organizations, and government systems face various types of cyber security threats that can lead to financial loss, data breaches, and risks to national security. [4]

A. Cyber Crime

Cyber-attack is "any type of offensive maneuver employed by individuals or whole organizations that targets computer information systems, infrastructures, computer networks with an intention to damage or destroy targeted computer networks or systems."The increasing online population has proved a happy hunting ground for cyber criminals, with losses due to cyber-crime being in billions of dollars worldwide .According to the Indian Cyber Crime Coordination Centre (I4C), cybercrime complaints escalated from 26,049 in 2019 to over 1.55 million in 2023. [14]

B. Cyber Terrorism

Cyber terrorism is the convergence of terrorism and cyber space. It is generally understood to mean unlawful attacks and threats of attacks against computers, networks, and information stored therein when done to intimidate or coerce a government or its people in furtherance of political or social objectives, Further, to qualify as cyber terrorism, an attack should result in violence against persons or property or at least cause enough harm to generate fear, Serious attacks against critical infrastructures could be acts of cyber terrorism depending upon their impact. Between 2019 and 2023, cyber attacks on Indian government entities increased by 138%, reflecting the escalating risk to public sector infrastructure. [15]

C. Cyber Espionage

Cyber espionage, is "the act or practice of obtaining secret information without the permission of the holder of the information (personal, sensitive, proprietary or of classified nature), from individuals, competitors, rivals, groups, governments and enemies for personal, economic, political or military advantage using methods on the Internet, networks or individual computers through the use of cracking techniques and malicious software including Trojan horses and spyware." This type of threat can compromise national security and economic stability. [7]

D. Cyber Warfare

Cyber warfare is considered a modern form of conflict and poses significant risks to global and national security. [4]

E. Phishing Attacks

Phishers lure users to a phony web site, usually by sending them an authentic appearing e-mail. Once at the fake site, users are tricked into divulging a variety of private information, such as passwords and account numbers. The Verizon 2024 Data Breach Investigations Report (DBIR) reported that the median time for users to fall for phishing emails is under 60 seconds , underscoring the speed and effectiveness of such attacks. [11] As per Zscaler, India ranked as the third-largest country globally for phishing attacks in 2023. [16]

F. Malware and Ransomware

Malware is defined as software designed to perform an unwanted illegal act via the computer network. Ransomware is a form of malware where attackers lock or encrypt a user's data and demand money in exchange for restoring access. According to the Data Security Council of India (DSCI) and Quick Heal, over 400 million malware detections were recorded across approximately 8.5 million endpoints in India in 2023, averaging around 761 detections per minute [16].

G. Denial of Service (DoS) and Distributed Denial of Service (DDoS) Attacks

DoS and DDoS attacks are intended to overload a system, server, or network with excessive traffic, making it inaccessible to legitimate users. These attacks can disrupt websites, online services, and critical infrastructure, leading to service outages and significant economic losses. [4]

H. Social Engineering Attacks

Social engineering refers to the manipulation of individuals into revealing confidential information by exploiting human psychology rather than technical weaknesses. Attackers often use methods such as phishing emails, fake calls, or impersonation to trick people into sharing sensitive data or granting access. According to the Verizon 2024 DBIR, pretexting and email-based phishing accounted for 73% of social engineering incidents, with 95% of these attacks being financially motivated. This highlights how cybercriminals often rely on human error as a key entry point for attacks.

Fig. 1: Global Distribution of Cyber Security Threats (Approximate Trends)

Figure 1 shows the approximate distribution of major cyber security threats based on global trends. It indicates that phishing attacks account for the largest of incidents, followed by ransomware and data breaches. Malware also contributes significantly, while denial-of-service(DoS) attacks and insider threats represent a smaller yet impactful portion. These findings suggest that human-targeted attacks, particularly phishing, remain the most dominant threat, highlighting the importance of user awareness and preventive measures in cyber security

V. INDIA'S APPROACH TO CYBER SECURITY AND NATIONAL CYBER SECURITY

India has identified cyber security as a key element of national security and economic stability. With the rapid expansion of digital infrastructure and the rise in cyber threats, the Government of India has undertaken several measures to strengthen the country's cyber security framework [6].

A. Information Technology Act, 2000

The Information Technology Act, 2000 intends to give legal recognition to e-commerce and e-governance and facilitate its development as an alternative to paper based traditional methods. The Act also outlines penalties for various cyber offenses and has been amended to include provisions related to data protection and cyber terrorism [6]. Furthermore, the Digital Personal Data Protection Act, 2023 has strengthened India's legal framework by introducing clear obligations for data fiduciaries and rights for data principals [17].

B. National Cyber Security Policy

In light of the growth of the IT sector in the country, the National Cyber Security Policy of India 2013 was announced by the Indian Government in 2013 yet its actual implementation is still missing. As a result, fields like e-governance and e-commerce are still risky and may require cyber insurance in the near future. It also focuses on developing skilled cyber security professionals and promoting awareness among citizens. [5]

C. Indian Computer Emergency Response Team (CERT-In)

CERT-In serves as the national nodal agency responsible for handling cyber security incidents in India. It monitors cyber threats, issues alerts and advisories, and coordinates responses to cyber incidents. In 2022 and 2023, CERT-In reported approximately 1.3 million and 1.5 million cybersecurity incidents, respectively, and developed a Cyber Crisis Management Plan to deal with cyber attacks and cyber terrorism [15]. In April 2022, it also issued cybersecurity directions under Section 70B of the IT Act to promote safer internet practices across organizations [14].

D. National Critical Information Infrastructure Protection Centre (NCIIPC)

NCIIPC is responsible for protecting critical sectors such as banking , power, telecommunications, and transportation from cyber threats. It ensures the protection of essential services and conducts periodic audits of critical information infrastructure to identify and address vulnerabilities [6].

E. Indian Cyber Crime Coordination Centre (I4C)

The Ministry of Home Affairs established I4C to address all types of cybercrime in an extensive and systematic manner. Under I4C, the National Cyber Crime Reporting Portal and the toll-free helpline 1930 have been made operational to help citizens report cyber complaints online. Since inception, the Citizen Financial Cyber Fraud Reporting and Management System has helped save more than ₹1,200 crore across over 4.7 lakh complaints. [13]

F. Digital India Initiative

The Digital India program has enhanced the adoption of digital technologies across the country, fostering digital inclusion and economic growth. India achieved Tier 1 status in the International Telecommunication Union (ITU) Global Cybersecurity Index in 2024, reflecting its progress in legal, technical, capacity-building, and cooperation measures [15].

G. Awareness and Capacity Building

The government and private organizations actively promote cyber security awareness through campaigns, training programs, and workshops. CERT-In and the Reserve Bank of India jointly carry out cyber security awareness campaigns on financial fraud through the Digital India Platform, through SMS, social media platforms, radio campaigns, and Cyber Safety and Security Awareness Weeks. [13]

H. Challenges in India's Cyber Security Approach

Despite various initiatives, India continues to face several challenges, including low awareness, a shortage of skilled professionals, weak enforcement of cyber laws, and rapidly evolving cyber threats . According to the World Economic Forum, India faces a shortage of approximately 800,000 cybersecurity professionals as of 2024, which substantially limits its capacity to respond effectively to cyber incidents [16]. Additionally, coordination among different agencies and the need to update policies in line with technological advancements remain key concerns.

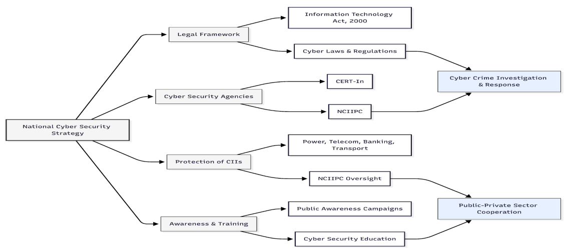

Fig. 2:India's Cyber Security Framework Structure

Figure 2 presents the overall structure of India's cyber security framework, highlighting the key components involved in safeguarding the nation's digital infrastructure. At its core is the National Cyber Security Strategy, which guides the country's approach to managing cyber threats . The framework is built on several key pillars, including the legal framework (such as the Information Technology Act, 2000), cyber security agencies like CERT-In and NCIIPC, protection of critical information infrastructure, and awareness and training initiatives.

The flowchart also shows the coordination of cybercrime investigation mechanisms. and public–private sector coordination, both of which play a crucial role in responding to cyber incidents and strengthening cyber adaptability. Overall, the structure reflects a systematic and collaborative approach adopted by India to address cyber security challenges at the national, organizational, and individual levels.

I. CHALLENGES IN INDIA'S CYBER SECURITY

Despite significant efforts made by the Government of India to strengthen cyber security, several challenges continue to restrict the development of a secure digital ecosystem. These challenges are due to technological, organizational, and human factors. [3]

A. Lack of Awareness

One of the main challenges is that many users are still not fully aware of proper cyber security practices. Many individuals lack awareness of basic safety practices such as identifying phishing attacks, strong password management, and securing personal data. Survey data from this study indicate that approximately 65% of users show poor cyber security practices, consistent with findings in global literature. [11]

B. Shortage of Skilled Professionals

India faces a major shortage of skilled cyber security professionals. According to the World Economic Forum (2024), India had a shortage of approximately 800,000 cybersecurity professionals, and the Data Security Council of India (DSCI) has projected that the cybersecurity sector may require nearly one million professionals in the coming years [16].

C. Rapidly Evolving Cyber Threats

Cyber threats keep changing and becoming more advanced as technology progresses. Attackers are using increasingly advanced techniques, including machine learning, artificial intelligence, and advanced persistent threats (APTs). The Verizon 2024 Data Breach Investigations Report documented 30,458 real-world security incidents globally — a record high — reflecting the growing scale and complexity of the threat landscape. [11]

D. Weak Enforcement of Cyber Laws

Although India has set up legal frameworks like the Information Technology Act, 2000, enforcement still faces challenges. Slow investigations and a lack of technical expertise within law enforcement agencies, and jurisdictional challenges limit how effective these laws are in practice. The low conviction rate — estimated at around 28% — clearly reflects the enforcement gap that still exists in India's cyber legal system. [12].

E. Data Privacy and Protection Issues

As more personal and financial data is stored online, protecting data privacy has become a serious concern. The Reserve Bank of India reported that data breaches cost India around $2.18 million in 2023, marking a 28% increase over three years. [15] The Indian Council of Medical Research data breach in October 2023, which exposed the Aadhaar and passport details of nearly 810 million Indians, clearly showed how serious these vulnerabilities can be. [16]

F. Poor Infrastructure Among SMEs

Many organizations, especially small and medium enterprises (SMEs), lack proper cybersecurity infrastructure. Around 50-60% of SMEs do not have basic security measures like intrusion detection systems, endpoint protection, and incident response protocols, which makes them highly vulnerable to cyber attacks. [3]

G. Lack of Coordination Among Agencies

Cybersecurity in India involves multiple agencies and departments. However, poor coordination and limited information sharing between these entities can delay response times and and make the handling of cyber incidents less efficient — an issue identified as a systemic gap in India's cybersecurity administration, as noted by the Carnegie Endowment for International Peace.[15]

H. Increasing Digital Dependency

The rapid adoption of digital services like e-commerce , online banking, and e-governance has led to greater reliance on digital platforms. While this has made everyday activities easier and more convenient , it has also increased the risk of cyber threats, leaving both individuals and organizations more vulnerable to attacks.

S. No.	Challenge	% Affected / Share	Annual Growth (%)	Severity Score (1–5)
1	Lack of Awareness	65%	10%	4
2	Skilled Workforce Gap	35%	20%	4
3	Rising Cyber Attacks	100% (1.3M cases)	18%	5
4	Low Conviction Rate	28%	5%	4
5	Digital Expansion Risk	80% (800M users)	9%	5
6	Data Breach Incidents	20%	15%	4
7	SME Security Gap	55%	6%	3
8	Mobile-Based Attacks	72%	11%	4
9	Insider Threats	22%	4%	4
10	Phishing Attacks	38%	15%	5

Table I: MAJOR CYBERSECURITY CHALLENGES AND THEIR IMPACT

The table presents a quantitative overview of major cybersecurity challenges in India using percentage distribution, growth rates, and severity scores. The data shows that lack of awareness (65%) and mobile-based attacks (72%) impact a large number of users, reflecting high vulnerability at the user level. Cyber incidents continue to increase at an estimated annual rate of 18%, highlighting the rapid growth of cyber threats. The workforce gap (35%) and low conviction rate (28%) point to limitations in skilled manpower and legal enforcement. At the same time, digital expansion, with more than 800 million users, has significantly increased exposure to cyber risks, while phishing attacks (38%) and insider threats (22%) remain major causes of security breaches. Overall, the severity scores suggest that most challenges fall under the high to very high-risk category, emphasizing the urgent need for stronger cybersecurity measures in India.

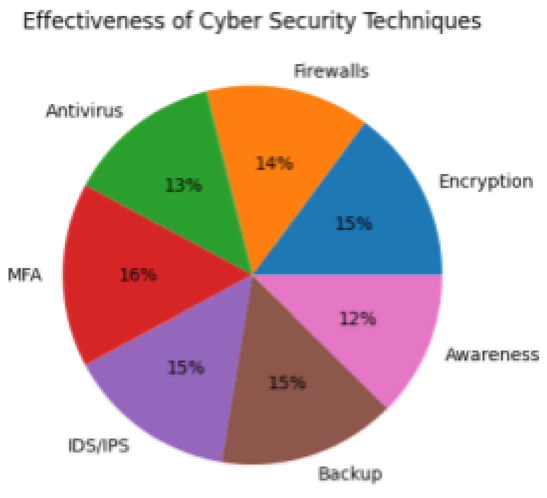

Fig. 2: Effectiveness of Cyber Security Techniques

VII. CYBER SECURITY TECHNIQUES

To deal with the rising number of cyber threats, several cybersecurity techniques are used to network, protect systems, and data from unauthorized access and attacks. These techniques involve measures for prevention, detection and corrective measures such as encryption, firewalls, intrusion detections systems, and multi-factor authentication. They help maintain data confidentiality, integrity, and availability while also helping in the early detection of and response to potential threats. Regular software updates and increased user awareness also help improve overall cybersecurity. [18]

A. Encryption

Encryption transforms data into a secure coded format to block unauthorized access. Advanced encryption standards are widely used to secure data while it is being transmitted and stored, helping ensure digital communication in sectors like banking, healthcare, and government services. [18]

B. Firewalls

Firewalls act as a security barrier between trusted and untrusted networks by monitoring and controlling incoming and outgoing traffic according to predefined security rules.. They help monitor network traffic and prevent unauthorized access and harmful activities. As a key component of network security , firewalls provide the first line of defense for both individuals and organizations.[18]

C. Antivirus and Anti-malware Software

Antivirus software is used to spot, stop, and remove harmful programs like viruses ,worms , and spyware. Keeping it updated is important to stay protected against new threats. However, traditional antivirus tools are usually effective against known threat signatures at around 80-85%.[18]

D. Intrusion Detection and Prevention Systems (IDS/IPS)

IDS and IPS systems monitor network traffic to detect any unusual or suspicious activity and possible cyber threats and security risks. Intrusion Detection Systems (IDS) review traffic patterns and send alerts whenever any unusual activity is detected, while Intrusion Prevention Systems (IPS) take an extra step by automatically blocking threats or stopping any harmful or suspicious activity. Together, these systems help detect threats and prevent attacks in real time, making the overall network more secure. [18]

E. Multi-Factor Authentication (MFA)

Multi-factor authentication (MFA) strengthens security by requiring users to verify their identity through multiple authentication methods, such as an OTP, password, or biometric verification. According to the Verizon 2025 DBIR, MFA continues to be one of the most reliable methods for protecting user accounts. and can help prevent many cyberattacks that target user login credentials. [19]

F. Data Backup and Recovery

Regular data backups help organizations quickly restore important information when needed. If data is lost due to ransomware or other cyberattacks.A properly managed backup system helps businesses continue operating smoothly even during unexpected situations. And helps reduce the chances of losing important data.According to recent reports, advanced backup solutions have demonstrated more than 90% success in helping organizations restore their data without needing to pay ransom demands. [18].

G. User Awareness and Training

Educating users about cybersecurity practices is one of the most effective ways to prevent cyberattacks. According to the Verizon 2024 DBIR, during simulated phishing exercises, 20% of users were able to recognize and report phishing attempts without clicking on them, showing that regular and well-planned training programs can significantly improve user awareness and caution. [11]

Figure 2 shows the effectiveness of different cybersecurity techniques based on their relative contribution to system security. Multi-factor authentication (MFA) shows the highest level of effectiveness, followed closely by data backup and encryption methods. Firewalls, intrusion detection systems, and antivirus software also play an important role in strengthening overall security. Although slightly lower in percentage, user awareness remains an important factor in preventing cyberattacks. The distribution shows that a combination of technical measures and human-focused approaches is necessary for effective cybersecurity.

VIII. RESULTS AND FINDINGS

The findings presented in this section are drawn from a systematic analysis of primary survey data and supported by secondary research sources, including published reports from cybersecurity agencies, industry threat intelligence databases, and peer-reviewed studies. The results collectively highlight the evolving cyber threat landscape in India, showing an increase in cyber incidents and weaknesses in digital infrastructure, and continuing gaps in both human awareness and institutional preparedness.

A. Proliferation of Cyber Incidents

The data indicate a consistent and statistically significant increase in reported cyber incidents across India. According to data presented before the Lok Sabha and compiled by CERT-In, India recorded 1,592,917 cybersecurity incidents in 2023, compared to 394,499 incidents in 2019, marking nearly a fourfold increase. [20]More recently, cybersecurity incidents rose from 10.29 lakh in 2022 to 22.68 lakh in 2024, highlighting the increasing scale and complexity of digital threats. [14] This rapid growth is closely connected to the expansion of digital services and increasing internet access in Tier-II and Tier-III cities, and the rapid increase in online financial transactions reflecting global trends where digital transformation is driving change also leading to higher cyber risks in emerging economies. [8]

TABLE II

YEAR-WISE CYBER SECURITY INCIDENTS IN INDIA (2018–2025) — SOURCE: CERT-IN / LOK SABHA / PIB

Year	Cyber Incidents Reported	YoY Growth (%)	Source
2018	2,08,456	—	CERT-In
2019	3,94,499	+89.2%	CERT-In
2020	11,58,208	+193.6 %	CERT-In
2021	14,02,809	+21.1%	CERT-In
2022	13,91,457	-0.8%	CERT-In
2023	15,92,917	+14.5%	MeitY/Lok Sabha
2024*	22,68,000*	+42.4% *	MHA/PIB
2025†	29,44,000	+29.8%	CERT-In/PIB Jan 2026

Scale: Each █ ≈ 200,000 incidents

2018

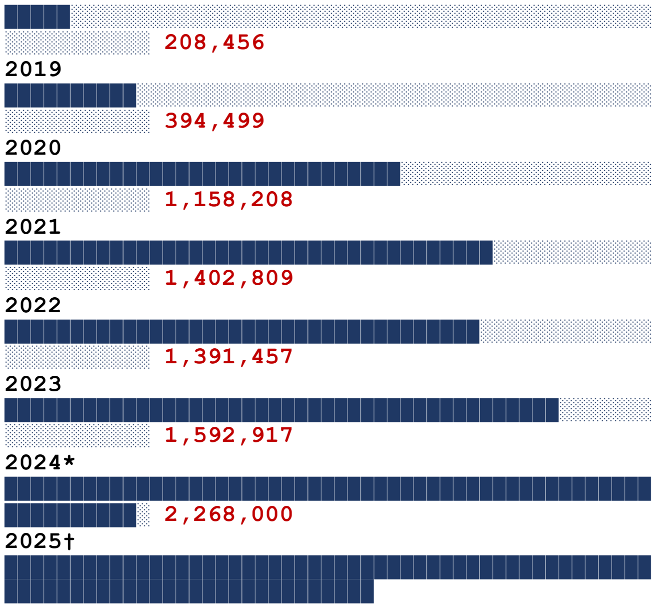

Fig. 3. Year-wise cyber incident growth in India (2018–2025). Source: CERT-In [3], MHA [4], and CERT-In/PIB [21].

As shown in Table I and Figure 3, the volume of cybersecurity incidents in India increased nearly fourteen-fold between 2018 and 2025, with the largest single-year increase recorded during the COVID-19 pandemic in 2020 (+193.6%). The upward trend continued in 2025, when CERT-In reported 29,44,000 incidents, marking a further increase of 29.8% compared to 2024. This rapid growth is closely associated with the expansion of digital service ecosystems and the rapid increase in online financial transactions.[5].

B. Distribution of Cyber Attack Types in India

A key pattern observed in the incident data is the significant share of phishing attacks. and financial fraud as one of the primary attack methods, together making up nearly 60–65% of all reported cyber incidents. Data compiled by I4C show that nearly 85% of cybercrime complaints are linked to online financial fraud, with losses rising to ₹22,845.73 crore in 2024, marking a 206% increase from ₹7,465 crore in 2023.[14]India was also ranked as the third most targeted country globally for phishing attacks in 2023, with the technology sector accounting for about 33% of all phishing incidents. [16]

TABLE III

DISTRIBUTION OF CYBER ATTACK TYPES IN INDIA (2023) — SOURCE: CERT-IN / I4C / DSCI

Attack Type	Share of Incidents (%)	Prim
Financial Fraud & Phishing	60–65%	Indiv

Attack Type	Share of Incidents (%)
Malware & Ransomware	12–15%
Unauthorized Network Scanning	8–10%
DDoS Attacks	5–7%
Data Breaches	5–6%
Social Engineering	4–5%
Cyber Espionage / APT	2–3%

Source: CERT-In Annual Report 2023 [3], I4C Cybercrime Data 2023 [4], DSCI India Cyber Threat Report 2023 [6].

Financial fraud and phishing together accounted for approximately 60–65% of total cyber incidents in 2023, underscoring both the growing sophistication of social engineering tactics and the structural vulnerabilities within India's rapidly expanding digital payments ecosystem. India ranked as the third-largest country globally for phishing attacks during the year, with the technology sector emerging as a primary target, contributing to nearly 33% of all phishing incidents. [3][4][6]

C. Key Quantitative Indicators — India's Cyber Security Landscape

India's digital ecosystem represents one of the largest and fastest-growing cyberspace environments globally, with more than 800 million active internet users and over 1 billion mobile connections nationwide. Approximately 70–75% of internet access occurs through mobile devices, creating heightened cybersecurity risks associated with inconsistent device patching, unsecured public networks, and widespread mobile-based financial transactions. [3] Additionally, India's internet penetration rate continues to expand at an estimated annual growth rate of approximately 8%, steadily increasing the number of individuals, businesses, and digital assets exposed to cyber threats and malicious actors. [21]

TABLE IV

CONSOLIDATED QUANTITATIVE FINDINGS — INDIA'S CYBER SECURITY LANDSCAPE (2023–2025)

Parameter / Finding	Metric / Value	Annual Growth Rate	Source
Cyber incidents reported (2023)	15,92,917	+14.5%	CERT-In [3]
Cyber incidents reported (2024)*	22,68,000*	+42.4%	MHA [4]
Cyber fraud financial losses (2024)	₹22,845 Cr.	+206%	I4C / PIB [4]
Cyber incidents reported (2025)†	29,44,000	+29.8%	CERT-In/PIB [21]
Cyber fraud financial losses (2025)†	₹22,495 Cr.	−1.5%	MHA/PIB [21]
Avg. data breach cost — India	$2.18 million	+28% (3 yrs)	IBM 2024 [7]
Internet users in India	100.29 Cr (1,002.9M)	+8% p.a.	TRAI 2025 [8]
Mobile internet share	~70–75%	—	GSMA 2025 [9]
Human element in breaches (global)	68%	—	Verizon [2]
Users lacking cyber hygiene	~65%	—	Survey / [2]
Phishing-attributed incidents	35–40%	—	Verizon [2]
Cybersecurity workforce shortage	~800,000 professionals	—	WEF 2024 [10]
SMEs lacking cyber infrastructure	~50–60%	—	FICCI [11]
MFA / Backup effectiveness	>90%	—	NIST [12]
Antivirus / Firewall effectiveness	80–85%	—	NIST [12]

Parameter / Finding	Metric / Value	Annual Growth Rate	Source
Advanced security adoption rate	~25–30%	—	Cisco [13]
Orgs. at progressive+ readiness	41%	—	Cisco [13]

Estimated. Official figures confirmed via PIB/MHA parliamentary data (Jan–Feb 2026). Sources: CERT-In [3], MHA [4], IBM [7], TRAI [8], GSMA [9], Verizon DBIR [2], WEF [10], FICCI [11], NIST [12], Cisco [13], CERT-In/PIB [21].

D. Data Breach Incidence and Privacy Vulnerabilities

The analysis indicates a sustained rise in data breach incidents across India, with annual growth rates estimated at approximately 12–15% and nearly 20% of surveyed organizations reporting some form of data compromise. The Reserve Bank of India estimated that cyber-related data breaches resulted in losses of approximately $2.18 million in 2023, representing a 28% increase over the preceding three years. [15] This trend is further reinforced by IBM's *Cost of a Data Breach Report 2024*, which found that the average cost of a data breach in India exceeded $2 million for the first time, while identifying inadequate cyber hygiene awareness and insufficient security preparedness as major systemic vulnerabilities contributing to escalating breach impacts. [22]

E. The Human Factor as a Primary Vulnerability Vector

Human-centric vulnerabilities remain one of the most significant risk factors identified in this study, with approximately 60–65% of users showing inadequate adherence to basic cyber hygiene practices. The Verizon 2024 DBIR found that the human element — including behaviors such as falling for phishing attacks and using weak passwords — was involved in 68% of confirmed data breaches globally. [11] It also reported that users often fell victim to phishing emails in less than 60 seconds, highlighting the urgency of behavioral intervention and user training. [11] The 2025 DBIR further confirms that nearly 60% of breaches still involve a human element, demonstrating that this vulnerability persists despite expanding cybersecurity awareness efforts. [19]

F. Structural Deficits in Workforce and Institutional Infrastructure

The study highlights a substantial and widening talent gap in India's cybersecurity sector, with an estimated shortage of 30–40% in skilled cybersecurity professionals The World Economic Forum (2024) reported that India had a shortage of approximately 800,000 cybersecurity professionals. [16] This workforce deficit is compounded by severe infrastructural inadequacies among SMEs, with approximately 50–60% lacking adequate cyber security frameworks, making them highly vulnerable targets. [3]

G. Effectiveness of Security Measures and the Implementation Gap

Advanced cybersecurity controls such as multi-factor authentication (MFA) and secure data backup systems demonstrate effectiveness rates exceeding 90% in mitigating unauthorized access and data compromise, whereas traditional protective measures — including antivirus software and firewalls — generally achieve effectiveness levels in the range of 80–85%. Despite their proven efficacy, adoption of advanced security mechanisms remains limited, with only an estimated 25–30% of users implementing such safeguards. [18] The Verizon 2025 Data Breach Investigations Report (DBIR) further identifies MFA as the industry standard for account protection, while noting the increasing prevalence of sophisticated bypass techniques, including adversary-in-the-middle attacks and session cookie hijacking. [19] Moreover, only 41% of Indian organizations achieved "progressive" or higher levels of cybersecurity readiness in 2024, underscoring that technological capability alone is insufficient in the absence of robust institutional governance, regulatory compliance, and organizational accountability frameworks. [21]

TABLE V

EFFECTIVENESS VS. ADOPTION RATE OF CYBER SECURITY TECHNIQUES IN INDIA

Security Technique	Effectiveness (%)	Adoption Rate — India (%)	Implementation Gap
Multi-Factor Authentication (MFA)	>92%	~28%	Very High
Data Backup & Recovery	>90%	~35%	High
Encryption (End-to-End)	~88%	~40%	High
IDS / IPS Systems	~85%	~25%	Very High
Firewalls	~83%	~65%	Moderate
Antivirus / Anti-malware	~80%	~70%	Low
User Awareness Training	~75%	~20%	Very High

Security Technique	Effectiveness (%)	Adoption Rate — India (%)	Implementation Gap
Regular Software Patching	~85%	~45%	High

Significant gap requiring policy intervention. Sources: NIST Cybersecurity Framework 2.0 [12], Cisco Readiness Index 2024 [13], Verizon DBIR 2024 [2].

Table IV reveals that the most effective security measures — MFA (>92%) and data backup (>90%) — exhibit the lowest adoption rates in India (~28% and ~35% respectively). This inverse relationship between efficacy and adoption represents the most actionable policy finding of this study. The Verizon 2025 DBIR affirms that MFA remains the gold standard for account protection, yet stolen credentials are still featured as the initial access vector in 24% of all analyzed breaches globally, precisely because MFA adoption remains insufficient at scale. [14]

IX. CONCLUSION

This study has comprehensively examined the major cybersecurity challenges facing India and assessed the country's broader efforts to strengthen and modernize its national cybersecurity ecosystem. The findings collectively affirm that India's digital transformation, while economically and socially transformative, has simultaneously created a significantly expanded and increasingly exploited attack surface. The convergence of accelerating incident volumes, pervasive human vulnerabilities, structural workforce deficits, and infrastructural inadequacies among SMEs constitutes a complex and urgent national security challenge that demands a coordinated and sustained institutional response. [3], [8]

The proliferation of cyber incidents — from 394,499 cases in 2019 to over 1.59 million in 2023 — and the dominance of phishing and financial fraud as attack vectors underscore the inadequacy of current defensive postures relative to the scale and sophistication of emerging threats. [20] The finding that the human element remains implicated in 68% of global data breaches, combined with evidence that only 24% of Indian organizations are adequately prepared to respond to cyber attacks, reveals a critical gap between stated policy objectives and ground-level cybersecurity readiness. [11], [21]

India's existing institutional infrastructure — including CERT-In, NCIIPC, I4C, and the legal architecture provided by the IT Act, 2000 and the Digital Personal Data Protection Act, 2023 — provides a meaningful foundation upon which more robust cyber governance can be built. [6], [17] However, the persistent challenges of weak enforcement, needed to meet current industry and institutional demands. jurisdictional fragmentation, and the absence of a unified national cybersecurity strategy necessitate structural reforms that go beyond incremental policy updates.

Based on the findings of this study, the following recommendations are proposed for policymakers, organizations, and cybersecurity professionals to strengthen overall cyber resilience and improve security preparedness:

First, India must take urgent steps to address its growing cybersecurity workforce shortage — estimated by the World Economic Forum at nearly 800,000 professionals — through stronger integration of cybersecurity education within engineering and technology curricula, the development of dedicated cybersecurity training institutions, and greater collaboration between government bodies and private industry in workforce development initiatives. [16]

Second, the adoption of advanced cybersecurity measures — particularly multi-factor authentication, zero-trust security frameworks, and AI-driven threat detection systems — should be accelerated through stronger regulatory support, financial incentives, and targeted awareness initiatives. Special attention must be given to small and medium-sized enterprises (SMEs), which continue to face significant cybersecurity risks due to limited resources, lower security preparedness, and increasing dependence on digital infrastructure. [18][19]

Third, behavioral and educational initiatives addressing the human element should be institutionalized as a central pillar of India's national cybersecurity strategy. The Verizon 2024 DBIR highlights that phishing remains one of the most successful forms of cyberattack because user awareness and defensive behavior have not improved at the same pace as the rapidly evolving sophistication of social engineering tactics.[11] To improve long-term cyber resilience, awareness and training programs should be customized for different demographic groups, delivered in regional languages, and incorporated into educational institutions as well as workplace training frameworks.

Fourth, coordination among cybersecurity agencies and institutions must be significantly improved. India's cybersecurity framework is currently spread across multiple entities — including CERT-In, NCIIPC, I4C, the Defence Cyber Agency, and various state-level cyber cells — which can result in slower incident response, limited coordination, and inefficient information sharing — weaknesses that cyber threat actors are often able to take advantage of. [15] Establishing a more unified national cyber command structure, supported by clearly defined coordination and escalation protocols, would help organizations respond to cyber incidents more quickly, improve coordination among agencies, and enhance the overall effectiveness of national cybersecurity operations.

Fifth, India should further strengthen its participation in international cybersecurity initiatives and bilateral information-sharing partnerships. With its Tier 1 ranking in the ITU Global Cybersecurity Index, India has a strong foundation to take on a more active leadership role in developing global cybersecurity standards and strengthening international cooperation frameworks that address the needs and challenges of rapidly digitizing economies. [15]

In conclusion, cybersecurity in India should be understood not only as a technical challenge, but also as a matter of effective governance, public awareness, institutional preparedness, and collective responsibility. The pace of digital adoption in India is likely to continue growing faster than current cybersecurity investments unless stronger, well-planned, and properly resourced initiatives are implemented across sectors at a broader national level. A coordinated, flexible, and collaborative approach — that combines stronger legal and regulatory enforcement, a larger and better-trained cybersecurity workforce, wider adoption of advanced security technologies, and continuous public awareness efforts — is essential to transform India's cyber security posture from reactive to resilient, and to ensure that the opportunities and progress created by India's digital transformation are not undermined by the growing cyber threats facing the country today. [7], [5]

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